

DBMS Assignment 9

Instructions:

- Submit one .pdf and one .ipynb file.
 - The pdf file should contain the query and outputs of PGAdmin/psql terminal outputs along with explanations
 - All the python codes should be in one .ipynb (Jupyter Notebook) file
 - Ensure that the code is neat and commented.
 - The name should be <roll no>_Assignment9.ipynb/.pdf
 - Total Marks = 30
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Use DVD Rental database.

Q1)

- a) Write a query to retrieve all payments strictly greater than \$5 but less than \$8, along with customer details. Also display results sorted by amount in descending order. (2)
- b) Create all suitable indexes and explain which index will also help with sorting. (3)

Q2)

- a) Write a query to find actors whose **last name starts with 'W'** and first name contains '**e'** **anywhere**'. Note the execution time and compare it the results after creating suitable indexes below. (1)
- b) Create separate indexes on the first name and last name of the actors and compare the performance. (2)
- c) Try to improve the execution time further by making a new modified index. (2)

Q3) Consider the following queries on the customer and film table from the dvd rental database:

- a) Find the details of the customer with the email 'melissa.king@sakilacustomer.org'. (1)
- b) Find all films which have a replacement cost greater than 3. (1)
- c) Create an appropriate index on the search keys to make the above queries efficient. Justify your index choice by comparing the execution time of the above queries with and without using index. (3)

Q4) For the dvd rental database:-

- a) Write a query to find all payments where the amount is exactly 4.99 and the payment_date falls between '2007-02-01' and '2007-02-15' .(1)
- b) Run the query in Python and analyze the time taken without any index. (1)

- c) Add a suitable index on amount and payment_date as well and compare the execution times with and without any index? (3)

Q5)

- a) Create a generalized inverted index (GIN) on the special features column of the film table. Compare the response time of the following query for both cases, with and without using index : (2)
- b) Find all films with the special features including "Behind the Scenes". (3)

Hint : Special Features is a string array type column,

anyarray @> anyarray → boolean

Does the first array contain the second, that is, does each element appearing in the second array equal some element of the first array? (Duplicates are not treated

specially, thus ARRAY[1] and ARRAY[1,1] are each considered to contain the other.)

ARRAY[1,4,3] @> ARRAY[3,1,3] → true

Q6)

- a) Create a table called 'dummy_data' with 2 attributes: 'text_data' and 'num_data'. 'text_data' contains randomly generated text sequences, and 'num_data' contains random integer values in the range 0-100. Populate the table with 100,000 records. (1)
- b)
 - 1. Write a query to filter records having 'ed' as a subsequence in 'text_data' and the value of 'num_data' > 50. (1)
 - 2. Analyze the execution time of the query without indexing. (1)
 - 3. Use appropriate indexing for reducing the execution time of the above query. Write a short justification for the choice of index. (2)